Fluid Dynamics II

Notes made by Finley Cooper

12th October 2026

Contents

1 Introduction **3**

2 Equations for a Newtonian Viscious Flow **3**

2.1 Dynamic viscosity 3

2.2 Rate of strain tensor 3

2.3 The stress tensor 4

2.4 Constitutive Equation for a Newtonian Fluid 5

1 Introduction

The Part IB Fluid Dynamics course mainly covers inviscid flow and irrotational flow including potential flow. The course covered an introduction to viscious flow for the very simple case where $\mathbf{u} = (u(y,t), 0)$ in 2D parallel flow.

This course is about the effects of viscosity and answers the following very important questions

(i) How does viscosity affect the equations for a general flow?

(ii) What happens if viscosity dominates over the inertia of the fluid?

(iii) What happens if the flow is nearly paralllel?

(iv) What happens if viscosity is small compared to inertia?

(v) What about stability?

2 Equations for a Newtonian Viscious Flow

2.1 Dynamic viscosity

Recall from Part IB Fluid Dynamics we have Couette flow where a plate at rest and a plate with velocity U are parallel with a space h between them filled with a viscous fluid. For a simple fluid we find that the force required to move the top plate F is proportional to $\frac{uA}{h}$. Hence the shear stress is

$$\tau_s = \frac{F}{A} = \mu \frac{U}{h}$$

where μ is the dynamic viscosity. The shear stress is due to the relative motion between each layer of fluid and its neighbours above and below.

2.2 Rate of strain tensor

Consider the velocity near a fixed point $\mathbf{0}$ *wlog.* Taylor expand $\mathbf{u}$ so we have the multivaraible Taylor expansion

$$u_i(\mathbf{x}) = u_i(0) + x_j \left.\frac{\partial u_i}{\partial x_j}\right|_{\mathbf{0}} + \frac{1}{2} x_j x_k \left.\frac{\partial^2 u_i}{\partial x_j \partial x_k}\right|_{\mathbf{0}} + \cdots$$

We drop the quadratic terms and decompose $\frac{\partial u_i}{\partial x_j}$ into symmetric and antisymmetric parts so

$$\begin{aligned}\frac{\partial u_i}{\partial x_j} &= \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) \\ &= e_{ij} + \frac{1}{2}\varepsilon_{jik}\boldsymbol{\omega}_k\end{aligned}$$

where e_{ij} is the *rate of strain tensor* and $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity. Hence we have that

$$\mathbf{u}(\mathbf{x}) \approx \mathbf{u}(\mathbf{0}) + \mathbf{e} \cdot \mathbf{x} + \frac{1}{2}\boldsymbol{\omega} \times \mathbf{x}.$$

We have that $e_{ij} = e_{ji}$ by definition so $\mathbf{e}$ is real symmetic and hence has real orthogonal eigenvectors which are the principle axis of strain. We have that $e_{ii} = \nabla \cdot \mathbf{u} = 0$ so the eigenvalues sum to 0. The eigenvalues are called the principle rates of strain.

2.3 The stress tensor

Recall in mechanics we consider two types of forces:

(i) Body forces $\mathbf{f}$ that act per unit volume, such as gravity;

(ii) Surface forces $\boldsymbol{\tau}$ that act per unit area, for example pressure and viscious shear stress

Let $\boldsymbol{\tau}(\mathbf{n};\mathbf{x},t)\delta S$ denote the force exerted by the positive side on the negative side of the small area element with $\mathbf{n}$ pointing from the negative to positive direction.

For example we have that $\boldsymbol{\tau}(\mathbf{n}) = -p\mathbf{n}$ for an inviscid fluid.

Now let $\mathbf{n}\delta S$ be the sloping face of a small tetrahedron of size ℓ whose three other faces are aligned with the cooridnate planes. The force acting on the sloped face is $\boldsymbol{\tau}(\mathbf{n})\delta S$. The force on the other faces are $\boldsymbol{\tau}(-\mathbf{1})\delta S_1, \boldsymbol{\tau}(-\mathbf{2})\delta S_2, \boldsymbol{\tau}(-\mathbf{3})\delta S_3$. Consider the force balance, the body force and inertia are $O(\ell^3)$ and the surface forces $\boldsymbol{\tau}$ are $O(\ell^2)$, so their ratio is $O(\ell) \to 0$ as $\ell \to 0$. So the surface forces must balance themselves for small ℓ.

Note, pressure and viscous stresses balance inertia and gravity on the characteristic length scale so small ℓ means that ℓ is small compared to the characteristic length scale L.

Since forces balance we have that

$$\boldsymbol{\tau}(\mathbf{n})\delta S + \boldsymbol{\tau}(-\mathbf{1})\delta S_1 + \boldsymbol{\tau}(-\mathbf{2})S_2 + \boldsymbol{\tau}(-\mathbf{3})\delta S_3 = 0.$$

Now

$$\delta S_i = \mathbf{i}\cdot(\mathbf{n}\delta S) = n_i \delta S$$

and $\boldsymbol{\tau}(-\mathbf{1}) = -\boldsymbol{\tau}(\mathbf{1})$ by Newton's Third Law. Putting this in, we get that

$$\boldsymbol{\tau}(\mathbf{n}) = \boldsymbol{\tau}(\mathbf{1})n_1 + \boldsymbol{\tau}(\mathbf{2})n_2 + \boldsymbol{\tau}(\mathbf{3})n_3$$

which shows that the $\boldsymbol{\tau}$ is linear in $\mathbf{n}$. So we have that

$$\tau_i(\mathbf{n}) = \sigma_{ij}n_j$$

where $\sigma_{ij} = \tau_i(j)$.

Definition. (Stress tensor) We call $\sigma(\mathbf{x},t)$, as constructed above, the *stress tensor*.

The key result is that the stress $\boldsymbol{\tau}$ exerted on a small area element (by the outside on the inside) is given by $\boldsymbol{\tau}(\mathbf{n}) = \sigma\cdot\mathbf{n}$.

For the invisicid case σ is just the identity tensor.

We will now show that σ is a symmetric tensor.

Consider the angular momentum balance on a small ($\ell \ll L$) arbitrary volume V about a fixed point in V we'll call $\mathbf{0}$.

The moment of the surface forces is

$$\int_{\partial V} \varepsilon_{ijk}x_j\tau_{k\ell}n_\ell \mathrm{d}S$$

which is $O(\ell^3)$. Similarly the moment of the volume forces are inertia are

$$\int_V \mathbf{x} \times (\dots\dots) \mathrm{d}V$$

which also should be $O(\ell^4)$.

For small ℓ, the moments of the surface force must balance, hence

$$0 = \int_V \varepsilon_{ijk} \frac{\partial}{\partial x_\ell}(x_j \sigma_{k\ell}) \mathrm{d}V = \int_V \varepsilon_{ijk} \left(\sigma_{kj} + x_j \frac{\partial \sigma_{k\ell}}{\partial x_\ell} \right) \mathrm{d}V.$$

and as x_j is $O(\ell)$ we can take $\ell \to 0$ to get rid of the second term hence the integral is approximately

$$\sim \int_V \varepsilon_{ijk} \sigma_{kj} \mathrm{d}V.$$

But V is arbitrary so $\varepsilon_{ijk}\sigma_{kj} = 0$ hence $\sigma_{ij} = \sigma_{ji}$.

2.4 Constitutive Equation for a Newtonian Fluid

We divide the stress tensor into two terms:

(i) An isotropic pressure term $-p\delta_{ij}$ so $p = -\frac{1}{3}\sigma_{kk}$.

(ii) A deviatoric stress $\sigma_{ij}^{\mathrm{dev}}$ which is the deviation from isotropic.

So we have that

$$\sigma_{ij} = -p\delta_{ij} + \sigma_{ij}^{\mathrm{dev}}$$